

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – ResearchMind AI

Domain of Project – Education & Research | Artificial Intelligence (AI)

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Problem Statement - Research Agent

Problem Statement: ResearchMind AI – Agentic Research Assistant

The Challenge

Researchers, students, and professionals spend significant time searching for relevant literature, analyzing multiple research papers, managing citations, identifying research gaps, and synthesizing findings into meaningful reports. Information is scattered across journals, repositories, and documents, making the research process time-consuming and inefficient.

Traditional search systems only retrieve documents but do not provide deep understanding, cross-paper analysis, contradiction detection, or identification of unexplored research opportunities. This creates challenges in conducting high-quality literature reviews and discovering novel research directions.

The Objective

Build an **Agentic AI-powered Research Assistant** that automates the end-to-end research workflow through a collaborative multi-agent architecture. The solution should:

- **Research Planning** – Understand research questions and generate structured research plans.
- **Literature Retrieval (RAG)** – Search, retrieve, and summarize information from research papers, PDFs, and trusted sources.
- **Document Intelligence** – Extract, analyze, and organize knowledge from uploaded research documents.
- **Citation-Aware Research Generation** – Generate research reports with proper source attribution and references.
- **Research Gap Detection** – Identify underexplored areas, contradictions, limitations, and future research opportunities.
- **Knowledge Graph Generation** – Visualize relationships between concepts, methods, datasets, and findings.
- **Automated Research Proposal Generation** – Generate potential research topics, methodologies, and future work recommendations.
- **Multi-Agent System** –
 - Research Planner Agent** (research strategy generation)
 - Web Search Agent** (information retrieval)
 - Document Intelligence Agent** (PDF understanding & extraction)
 - RAG Retrieval Agent** (context retrieval)
 - Research Writer Agent** (report generation)
 - Citation Agent** (reference validation)
 - Research Gap Detection Agent** (opportunity discovery)
- **Interactive Research Dashboard** – Display research findings, citations, source analysis, knowledge graphs, and generated reports.

Technology Used

I used LangFlow + IBM watsonx.ai + Granite Models

AI & Agent Framework

- LangGraph (Multi-Agent Orchestration)
- RAG (Retrieval-Augmented Generation)
- IBM Granite Models
- Groq API (LLM Inference)

Data & Knowledge Layer

- ChromaDB (Vector Database)
- HuggingFace Embeddings

Backend

- FastAPI
- Supabase

Frontend

- Next.js 15
- Tailwind CSS
- ShadCN UI

Key Features Enabled

- Multi-Agent Research Workflow
- Semantic Search & Retrieval
- Citation-Aware Report Generation
- Research Gap Detection
- Automated Research Proposal Generation

Proposed Solution - ResearchMind AI

ResearchMind AI is a **multi-agent Agentic AI Research Assistant** built using **LangFlow, IBM watsonx.ai, IBM Granite Models, RAG, and LangGraph** to automate the complete research lifecycle.

The system enables users to:

- Search and retrieve relevant research papers and sources
- Analyze uploaded PDFs and extract key insights
- Generate citation-grounded literature reviews
- Detect research gaps, contradictions, and opportunities
- Build knowledge graphs connecting concepts and findings
- Generate novel research ideas and proposals
- Produce structured research reports with references

Multi-Agent Architecture

- **Research Planner Agent** – Creates research strategy
- **Web Search Agent** – Retrieves relevant sources
- **Document Intelligence Agent** – Processes PDFs and extracts knowledge
- **RAG Retrieval Agent** – Retrieves contextual information
- **Research Writer Agent** – Generates reports and summaries
- **Citation Agent** – Verifies and manages references
- **Research Gap Detector Agent** – Identifies unexplored opportunities

The platform transforms scattered research information into actionable insights, helping researchers save time, improve accuracy, and accelerate innovation.

Langflow component Used -

1) Chat Input

Accepts research queries, topics, and document upload requests from users.

2) IBM watsonx AI Agent

Processes user queries using IBM Granite models and coordinates research workflows.

3) IBM Granite Model

Granite-4.0-8B-Instruct → Used for research planning, summarization, report generation, and reasoning.

4) Chat Output

Displays literature reviews, research reports, citations, and research insights.

5) File Component

Uploads and processes research papers, PDFs, journals, and academic documents.

6) RAG Retrieval Component

Retrieves relevant information from the vector database to provide context-aware responses.

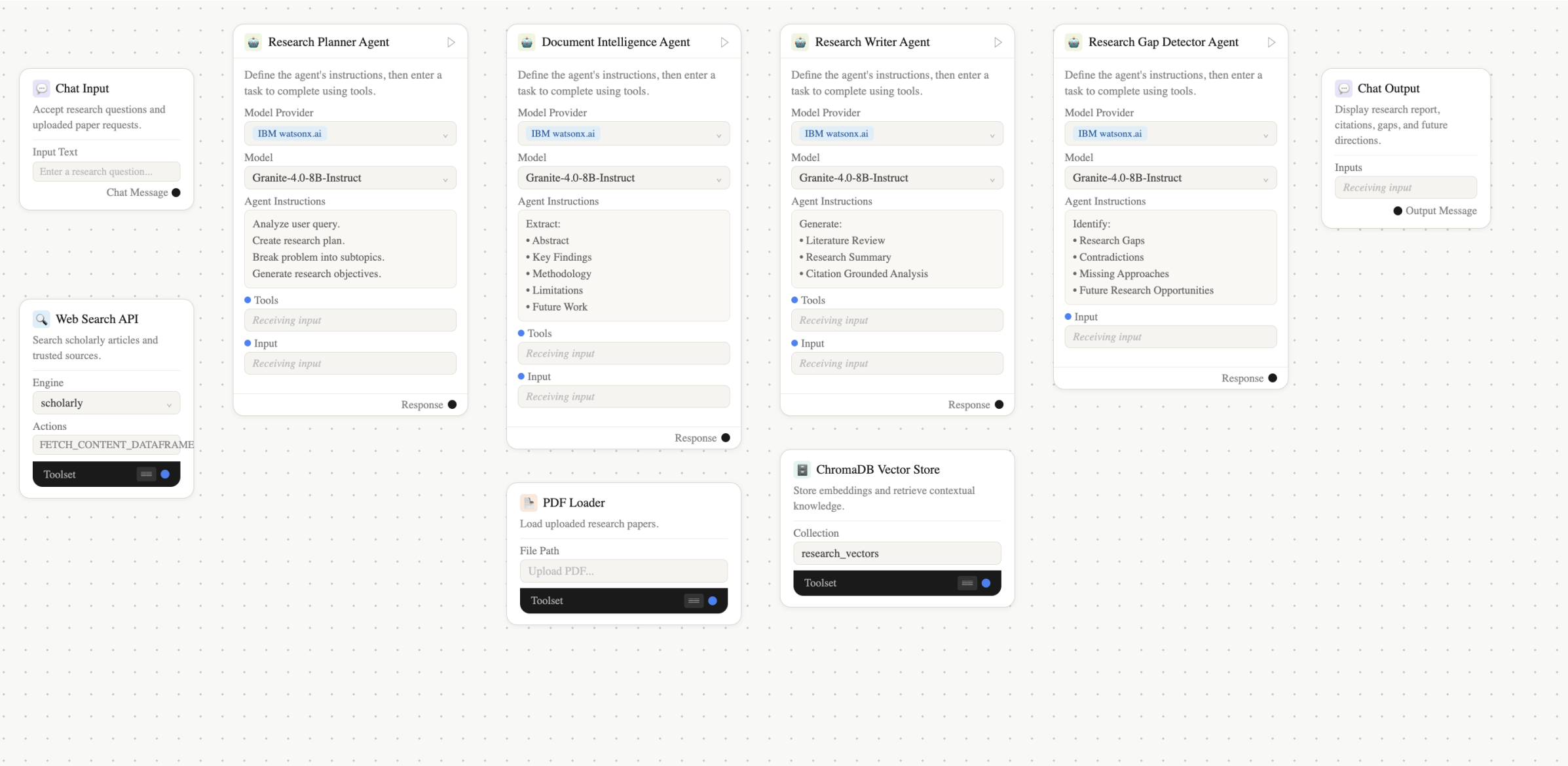
7) Vector Store (ChromaDB)

Stores document embeddings and enables semantic search across research papers.

8) Prompt Component

Manages specialized prompts for research planning, gap detection, and report generation.

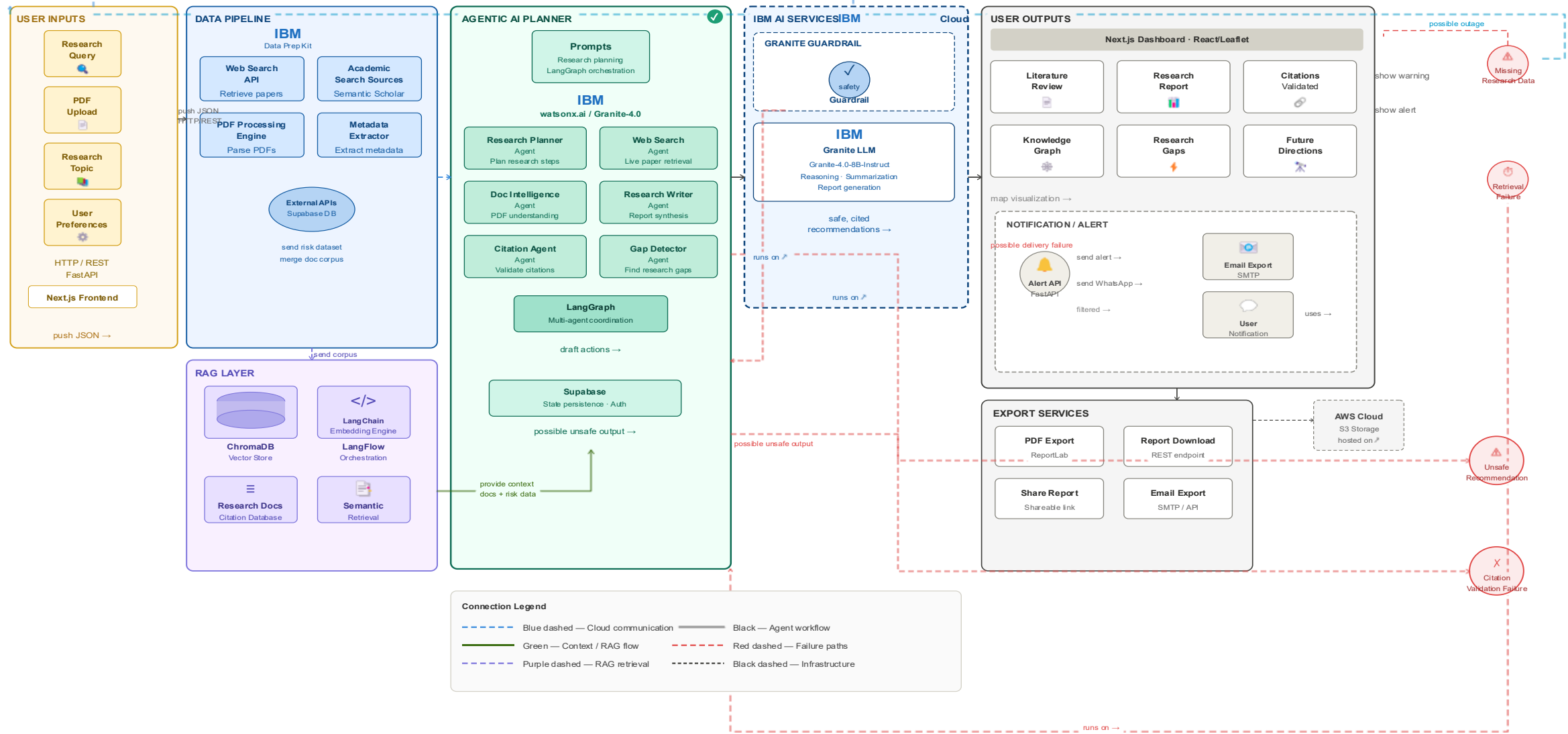
Langflow workflow -



Architecture blueprint

ResearchMind AI — Agentic AI Research Assistant Architecture

IBM watsonx.ai · IBM Granite · LangGraph · LangFlow · RAG · ChromaDB · FastAPI · Supabase · Next.js



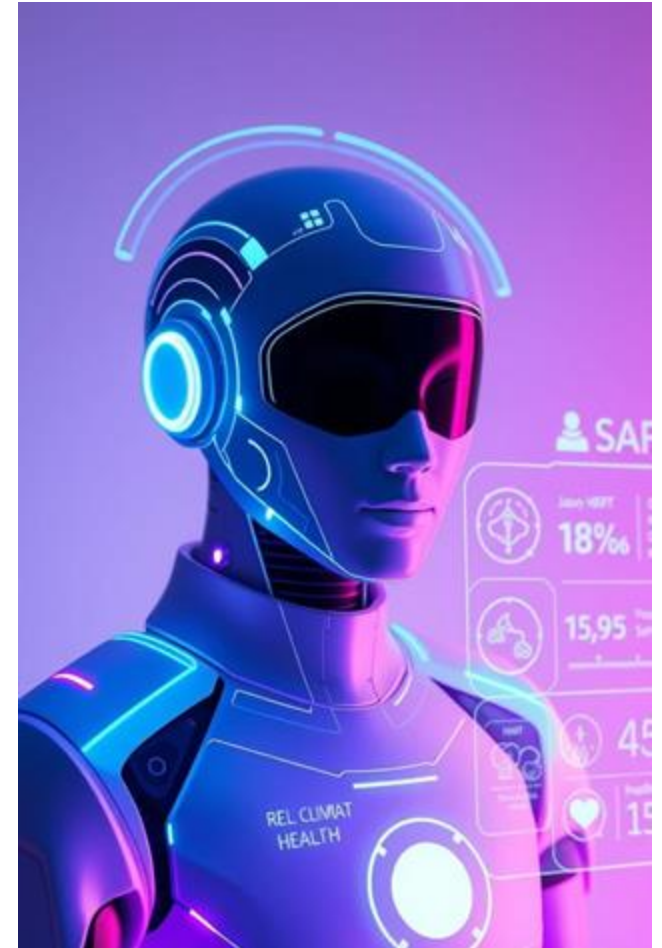
Role of Agentic AI in ResearchMind AI

Agentic AI enables ResearchMind AI to function as an intelligent and autonomous research assistant rather than a traditional search engine or chatbot. Using **IBM watsonx.ai** and **IBM Granite Models**, the system coordinates multiple specialized agents that collaborate to automate the entire research workflow. Each agent performs a dedicated task—such as planning research strategies, retrieving relevant literature, analyzing research papers, generating reports, managing citations, and detecting research gaps—while continuously sharing context with other agents to achieve a common objective.

The multi-agent architecture allows the system to break down complex research problems into smaller tasks and execute them efficiently. Agentic AI brings contextual understanding by analyzing user research goals, uploaded documents, and retrieved knowledge to generate highly relevant insights. It also supports dynamic decision-making, enabling agents to adapt their actions based on newly discovered information, citations, and research findings.

Additionally, agents communicate and collaborate throughout the workflow, ensuring accurate information retrieval, citation-grounded report generation, and identification of unexplored research opportunities. This coordinated approach reduces manual effort, improves research quality, and accelerates knowledge discovery.

Overall, Agentic AI makes ResearchMind AI proactive, context-aware, and goal-driven, acting as a **smart digital research companion** that assists researchers in literature review, research gap detection, knowledge synthesis, and innovation discovery.



Technology Used

- **LangFlow Platform** – For designing and orchestrating multi-agent research workflows visually.
- **IBM Granite Model (Granite-4.0-8B-Instruct)** – For research planning, reasoning, summarization, citation generation, and report creation.
- **IBM watsonx.ai** – For model deployment, inference, AI governance, and enterprise-grade AI services.
- **RAG (Retrieval-Augmented Generation)** – Retrieves trusted research information before generating responses, ensuring accurate and citation-grounded outputs.
- **ChromaDB Vector Database** – Stores document embeddings and enables semantic search across research papers and knowledge sources.
- **LangGraph** – Manages agent communication, workflow execution, and multi-agent orchestration.
- **FastAPI & Supabase** – Provide backend APIs, authentication, research session storage, and report management.
- **Next.js & Tailwind CSS** – Deliver a responsive and modern user interface for research interaction and visualization.

Novelty and Uniqueness

The proposed **ResearchMind AI** stands out by combining **Agentic AI, Multi-Agent Systems, RAG, Research Gap Detection, and Knowledge Graph Intelligence** into a single autonomous research platform.

Multi-Agent Intelligence – Instead of a traditional chatbot, multiple specialized agents collaborate to perform research planning, information retrieval, document analysis, report generation, and citation management.

Research Gap Detection – Automatically identifies unexplored areas, methodological limitations, contradictions, and future research opportunities from multiple research papers.

Citation-Grounded Research – Integrates RAG-based retrieval with AI generation to produce trustworthy insights supported by verifiable references.

Knowledge Graph Generation – Visualizes relationships between concepts, datasets, methods, and findings to improve research understanding.

Automated Research Proposal Generation – Transforms detected research gaps into potential research topics, methodologies, and innovation opportunities.

Document Intelligence – Analyzes uploaded PDFs and research papers to extract key findings, limitations, and future work recommendations.

Scalable & Enterprise-Ready – Built using IBM watsonx.ai, IBM Granite Models, LangFlow, and RAG architecture, ensuring reliability, scalability, and efficient research automation.

This makes the solution a **smart, adaptive, and proactive digital research companion**, not just a document search or question-answering system.

Git Hub Link :

<https://github.com/anuragpandey1rkt-cmyk/ResearchMind-AI>

Future Scope

1. Real-Time Academic Database Integration –

The system can be integrated with academic repositories such as IEEE Xplore, arXiv, Springer, ACM Digital Library, and Google Scholar to automatically retrieve the latest research papers and emerging trends. This will enable researchers to access up-to-date knowledge and conduct more comprehensive literature reviews.

2. AI-Powered Research Collaboration & Multilingual Support –

Future versions can support collaborative research workspaces where multiple researchers can share findings, annotations, and generated reports. Additionally, multilingual capabilities will allow users to analyze and generate research content in different languages, making the platform more accessible to the global research community.

3. Automated Research Proposal & Publication Assistant –

The platform can be enhanced to automatically generate complete research proposals, publication drafts, conference papers, and funding applications based on identified research gaps, significantly reducing the effort required in academic writing.

4. Advanced Knowledge Graph & Trend Prediction –

By leveraging large-scale research datasets and citation networks, the system can predict emerging research areas, identify high-impact topics, and visualize future technology trends to support strategic research planning.

Thank You!

Thank you for your time and interest.